

Multi-Cancer Early Detection Screening Report

Next-Generation Sequencing cfDNA Methylation Profiling (Synthetic Demonstration)

Patient Demographics

Name: Robert T. Sample
DOB: 11/23/1968 (Age 57)
Sex: Male
Patient ID: SYN-MRN-8849102

Ordering Clinician

Provider: Dr. Marcus Vance, MD
Facility: Bay Area Health System
NPI: 9982736450
Location: Fictional City, CA

Specimen Chain of Custody

Specimen ID: SYN-SPEC-MCED-4491
Specimen Type: Blood / Plasma
Tube: Streck cfDNA BCT (10.0 mL)
Collection Date: Aug 1, 2026
Received Date: Aug 2, 2026
Report Date: Aug 6, 2026

Result: Cancer Signal Not Detected

Status: Final
Assay: cfDNA Methylation NGS

Test Result Summary: No cancer signal was detected in this synthetic cell-free DNA (cfDNA) blood sample. Methylation patterns analyzed across target genomic regions are consistent with non-malignant profiles. This result does not completely rule out the presence of all types of malignancy.

Laboratory Observations & Biomarkers

Test / Analyte	LOINC	Result	Reference Range	Interpretation
Cancer Signal Status	94076-7	Cancer Signal Not Detected	Cancer Signal Not Detected	Normal
Predicted Cancer Signal Origin 1	94077-5	None Detected	None	Normal
cfDNA Concentration	Local	8.4 ng/mL	5.0 - 25.0 ng/mL	Normal
Bisulfite Conversion Efficiency	Local	99.8%	>= 99.0%	Normal
Sequencing Coverage Depth	Local	31,200x	>= 20,000x	Normal
Sequencing QC Pass Rate	Local	99.4%	>= 95.0%	Normal

Clinical Interpretation & Recommended Next Steps

A 'Cancer Signal Not Detected' result indicates that the targeted cell-free DNA methylation signatures associated with 50+ cancer types were not identified at actionable thresholds in this specimen. Recommended next steps include:

- Routine screening:** Continue age-appropriate standard preventative screening guidelines (such as colonoscopy, mammography, and prostate screening) as medically indicated.
- Follow-up:** Regular annual wellness examinations with primary care physician.

Methodology & Limitations

Methodology: Plasma cfDNA is extracted and treated with targeted bisulfite conversion followed by deep Next-Generation Sequencing (NGS) on Illumina NovaSeq 6000 systems. Epigenetic methylation patterns are evaluated using machine-learning classification models calibrated across multi-organ tissue datasets.

Limitations: A negative result does not eliminate the risk of cancer. Certain indolent tumors or cancers shedding low levels of cfDNA may not be detected.

Laboratory Director Authorization: Dr. Eleanor Hayes, MD, PhD, FCAP | Electronically Signed | Date: Aug 6, 2026 | CLIA ID: 00D9981245